

An Integrated GeoAI Framework for Instance-Level Disaster Impact Assessment Using Multimodal Remote Sensing and Demographic Exposure Modeling

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1 Introduction & Background

Natural hazards have increasing frequency and impact in recent decades, and disasters often inherently distribute uneven damages to impacted areas, amplifying pre-existing social inequities (Gaynor and Wilson, 2020; Lee et al., 2023). Thus, credible impact and justice analysis benefit from spatially resolved evidence at the scale of structures and households.

Within remote sensing for damage assessment, a large share of studies adopts one-stage, pixel-level semantic segmentation that predicts per-pixel damage from pre/post imagery, for example the xBD/xView2 ecosystem and follow-ons like PPM-SSNet or domain-adaptive semantic change detection (Su et al., 2020; Bai et al., 2020). In parallel, two-stage pipelines segment buildings first and then classify damage on building-centered crops, as in U-Net-based two-step models, ISPRS/xView2-inspired two-stage systems, and recent multi-disaster stacks (Alisjahbana et al., 2024; Deng and Wang, 2022). These approaches are effective for broad mapping and benchmark metrics, yet their outputs are not inherently building-oriented, which can reduce interpretability and reliability for downstream impact analysis in dense areas where adjacent roofs fuse or instance identity is ambiguous (Liao et al., 2021; Wagner et al., 2020). The remaining gap is consistent, per-building instances with polygons and stable IDs that support masked pooling, demographic coupling at fine scales, and explicit uncertainty reporting; the ideal solution first produces instance polygons, then attaches calibrated, per-building damage estimates suitable for equity-focused aggregation.

Complementing the academic literature, operational frameworks now let users run damage mapping for a chosen area and time, including cloud services that return multi-class damage layers from pre- and post-event imagery (Su et al., 2020). Humanitarian teams pair models with web maps to support field decisions, and national and UN programs offer area of interest (AOI)-driven rapid mapping at scale, such as UNOSAT’s AI-assisted flood mapping and dashboards (Gholami et al., 2022; UNITAR, 2020). Research groups also release event-focused applications and “integrated” stacks, including wildfire DamageMap and the DamageScope framework for post-disaster building damage (Galanis et al., 2021; Al Shafian et al., 2025). These systems improve accessibility and speed but typically output pixel masks or post-only layers, provide limited building-oriented products, and rarely include calibrated per-building probabilities or equity-aware demographic coupling. Motivated by these gaps,

we pursue a temporally leakage-aware end-to-end framework that produces pre-event instance footprints, calibrated building-level damage from paired pre- and post-event imagery, and scalable equity summaries with uncertainty from a simple AOI and time window.

Most end-to-end damage-mapping frameworks surface hard labels or heatmaps without publishing predictive uncertainty; examples include the xView2/xBD cloud demo, Microsoft AI for Good $\times$ 510 NLRC’s deployment, and UNOSAT’s AOI-driven rapid mapping, which emphasize speed and visualization rather than calibrated probabilities or confidence bands (Su et al., 2020; Gholami et al., 2022; UNITAR, 2020). At the model level, a newer line of work introduces uncertainty, for example BayeSiamMTL, a hybrid Bayesian deep model for building-damage assessment that outputs uncertainty-aware predictions (Hertel et al., 2025). In parallel, the segmentation literature shows that probability calibration, such as local temperature scaling, can materially reduce miscalibration (ECE) for dense prediction, yet such calibration is rarely reported in disaster-damage pipelines (Ding et al., 2021). Uncertainty handling is generally inconsistent across systems and seldom propagated to the building scale where decisions are made about specific structures, which limits informed decision support. Our framework addresses this by retaining per-building class-probability vectors, calibrating them, and propagating uncertainty at the building level, with optional aggregation to blocks or tracts via simple expectations or Monte Carlo when summaries are needed.

In summary, we position building instances as the core unit of analysis, adopt a leakage-aware two-stage design that separates footprint extraction from damage inference, and commit to calibrated probabilistic outputs that support equity-focused aggregation and decision-making. The remainder of the paper details the instance-extraction stage and shape regularization, the Siamese ordinal damage model with calibration, and the demographic coupling with uncertainty propagation, followed by evaluation on event-blocked splits and operational case studies.

2 Methodology

2.1 Dataset

2.1.1 2-Stage Model Dataset: the xBD / xView2 Dataset

The xView2 AI Challenge, hosted by the U.S. Defense Innovation Unit, targets two tasks from paired pre- and post-disaster satellite images: building localization and damage rating at the scene level, with the goal of reducing weeks of manual mapping to hours of inference. The underlying xBD dataset contains 850,736 annotated buildings across 45,362 km², with 9,168 paired 1024 $\times$ 1024 RGB training tiles spanning 19 disasters and five hazard types worldwide, with a very high spatial resolution of Ground Sample Distance (GSD) $\leq$ 0.8 m; in post-disaster images, each building receives a categorical damage label (Gupta et al., 2019; Alstad, 2020). The organizers note substantial variability across events and a strong class imbalance toward the “no damage” class, together with subtle visual distinctions between adjacent classes such as minor versus major damage, which together make the problem challenging (Alstad, 2020).

For each pre/post pair, challenge submissions produced two raster outputs: a building-localization PNG on the pre event image (binary per pixel) and a damage-classification PNG on the post event image with integer classes 0–4. Figure 1 provides an example set of pre-disaster, building footprint, post-disaster, and damage severity classes for one patch in the AOI in the dataset. Reports highlight that high building density degrades localization and that models often struggle to separate nearby severity classes, underscoring the need for architectures and training schemes that can handle dense urban scenes and ordered damage levels (Alstad, 2020).

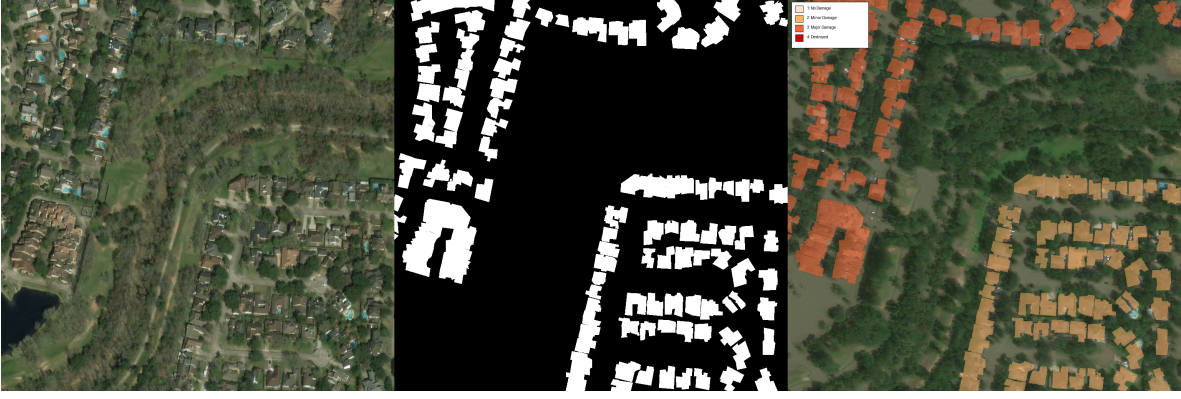


Figure 1: An example set of information in a given remote sensing cell in the xBD/xView2 dataset during Harvey 2017. The set for a given cell contains following images in 1024x1024 resolution: pre-disaster image (left), building polygons with ID's (middle), post-disaster image with polygon overlay, color-coded with damage severity labels (right).

2.1.2 Proxy Rule Dataset

The analysis integrates four core datasets: Harris County parcels with land-use classifications for building type, statewide Texas building footprints to capture building geometry, Census TIGER/Line CBG boundaries for geographic control, and ACS 5-year estimates for housing occupancy and people-per-unit ratios by structure type (tables B25032, B25033, and B25002 for occupied units, population by structure, and occupancy rates, respectively). For Harris County, the working inventory includes over 1.5 million parcels, roughly 10.6 million statewide building polygons from which county subsets are extracted.

2.2 End-to-End Framework

Figure 2 gives a conceptual visualization of how the framework aims to deliver end-to-end level disaster impact analysis. First, a 2-stage model will be trained on the xView2 dataset introduced in 2.1.1, aiming to eventually extract building instances and predict their corresponding disaster impact severity when migrated to unseen data, namely only the pre- and post-disaster imagery. Then, the framework would use occupancy unit proxy rules to assign demographic attributes to building instances so that eventually, this framework provides a building-level match between demographic and disaster impact.

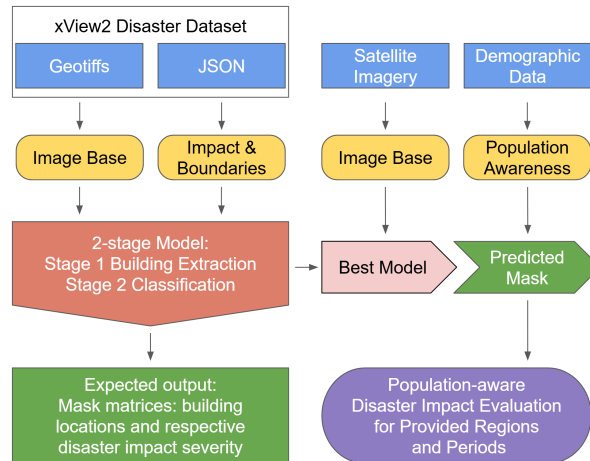


Figure 2: Conceptual overview of the proposed streamlined climate justice evaluation.

2.3 2-stage Model

2.3.1 Training and Inference Paradigms

Stage 1 is trained on pre-disaster tiles to predict building footprints; masks are cleaned and polygonized to produce per-building instances with stable IDs, visually presented by Figure 3. Stage 2 then consumes per-building, geometry-centered inputs: for each polygon, we generate paired pre/post crops (default 256×256 pixels), an instance mask M , and a ring/context mask $R = \text{buffer}(M, r) \setminus M$ with default ring radius $r=48$ pixels, visually presented by Figure 4 at a high level. The Stage 2 network is a Siamese encoder (default backbone `convnext_tiny`) that performs masked average pooling over M and R , forms pre/post feature deltas, and predicts ordinal damage with a CORAL head for $K=4$ severity levels. Training minimizes BCEWithLogits on the $K-1$ cumulative targets and evaluates macro-F1, per-class F1, Quadratic Weighted Kappa (QWK), Negative Log-Likelihood (NLL), and Expected Calibration Error (ECE) on a held-out validation set. To limit spatial/event leakage, splits are formed by `tile_id` rather than by row.

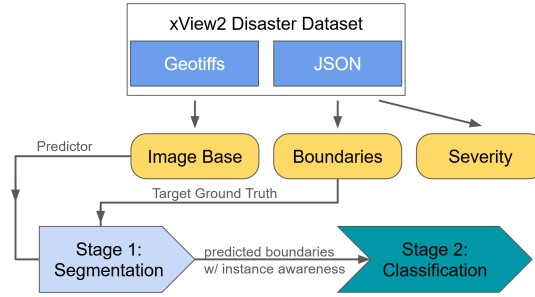


Figure 3: Stage 1 training pipeline.

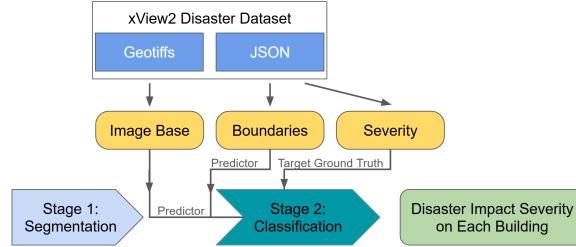


Figure 4: Stage 2 training pipeline.

Optimization uses AdamW with learning rate 2×10^{-4} , weight decay 0.05, automatic mixed precision, and a typical batch size of 16 samples per step. Checkpoints save both the best model by validation macro-F1 and the last epoch. The data loader applies ImageNet-style normalization, binarizes masks, and downsamples them to the feature resolution with nearest-neighbor to preserve boundaries. Because the label distribution is imbalanced (“no damage” dominates), an optional class-balanced sampler with tempered inverse-frequency weights is available; we additionally report a tolerant accuracy $\mathbb{I}(|\hat{y} - y| \leq 1)$ to reflect ordinal near-misses.

At inference on an unseen area, Stage 1 runs on pre-disaster imagery only to derive polygons. The same per-building profiles (pre/post crops, M , R) are then constructed and fed to Stage 2, which outputs a per-building probability vector $p \in [0, 1]^4$ and a hard class. Alongside the class, the system records confidence and uncertainty summaries (top probability, margin between the two most likely classes, entropy, and expected severity). Post-hoc calibration can be applied using either scalar temperature scaling in logit space or a class-wise vector temperature in probability space learned on validation events. Final outputs are stored per instance, rendered as overlays for visual QA, and made available

for downstream aggregation when regional summaries are required.

2.3.2 Stage 1: Segmentation

In Stage 1, we trained a U-Net model to perform binary semantic segmentation of building footprints from satellite imagery. Each input image was first preprocessed by resizing to 512×512 pixels and normalizing pixel intensities. Corresponding ground-truth masks were generated from polygon annotations, rasterized into binary images of the same size. The U-Net architecture consisted of an encoder–decoder structure with four downsampling and upsampling stages connected by skip connections. Each block in the encoder applied two 3×3 convolutions followed by batch normalization and ReLU activation, with max-pooling for downsampling. The bottleneck layer had 1024 channels. The decoder used transposed convolutions for upsampling, concatenating encoder features via skip connections before convolutional refinement. A final 1×1 convolution with sigmoid activation produced a per-pixel probability map indicating building presence.

For training, we used the Adam optimizer with an initial learning rate of 10^{-4} and binary cross-entropy loss. Data augmentation techniques such as random flips and rotations were applied to improve generalization. Models were trained for 50 epochs with a batch size of 8, and validation was performed after each epoch. At inference, predicted probability maps were thresholded at 0.5 to obtain final building masks.

2.3.3 Stage 1 to 2: Polygonization & Instance Profiling

Stage 1 building segmentation on pre-event tiles is converted into per-building instances for Stage 2 via a four-step, file-backed workflow. (1) From binarized building masks, inference applies a probability threshold, small-component removal, and optional morphological opening, writing one refined mask per tile. (2) Polygonization then extracts external contours, filters by minimum area, and simplifies vertex sequences, yielding an xBD-style JSON with WKT polygons—one feature per building instance. (3) A per-building index links each predicted polygon to its paired pre/post images and records tile dimensions and metadata in a single CSV (one row per instance) containing WKT geometry, image paths, and a unique building identifier. (4) Geometry-centered inputs for Stage 2 are created by cropping square windows around each polygon centroid from the pre- and post-event images, rasterizing an instance mask M from the polygon, and constructing a context ring $R = \text{buffer}(M, r) \setminus M$; paths to the crops and masks are written to an augmented samples CSV for training and inference.

This transition is parameterized to accommodate scene variability and operational constraints. Key parameters include the mask threshold, minimum area, and opening kernel for Stage 1 post-processing; polygon simplification tolerance for vectorization; and Stage 2 crop size and ring radius. Each step emits processed/skipped summaries for reproducibility, and end-to-end orchestrators stage case-study imagery into the expected naming scheme before invoking the full pipeline with auto-detected image size, resume/force options, optional limiting, and overlay generation for rapid visualization.

2.3.4 Stage 2: Multiclass Prediction, Uncertainty, Calibration

Damage is modeled as an ordinal $K=4$ classification problem using a Siamese backbone shared between pre- and post-event crops, and the model performs classification solely based on the two localized masks on the crops: the polygon mask and the spatially buffered mask. Let $f(\cdot)$ denote the shared encoder, $X^{\text{pre}}, X^{\text{post}}$ the input crops, and M, R the masks (downsampled to the feature map). Feature tensors are $F^{\text{pre}} = f(X^{\text{pre}})$ and $F^{\text{post}} = f(X^{\text{post}})$. Masked average pooling is applied over the interior and ring to obtain compact descriptors:

$$v(F, M) = \frac{\sum_u M_u F_u}{\sum_u M_u + \epsilon}, \quad v(F, R) = \frac{\sum_u R_u F_u}{\sum_u R_u + \epsilon},$$

forming $v^{\text{pre}}, v^{\text{post}}, v^{R,\text{pre}}, v^{R,\text{post}}$ and their change vectors $\Delta v = v^{\text{post}} - v^{\text{pre}}$ and $\Delta v^R = v^{R,\text{post}} - v^{R,\text{pre}}$. Figure 5 gives an example of how the profiling set for a building instance looks like.

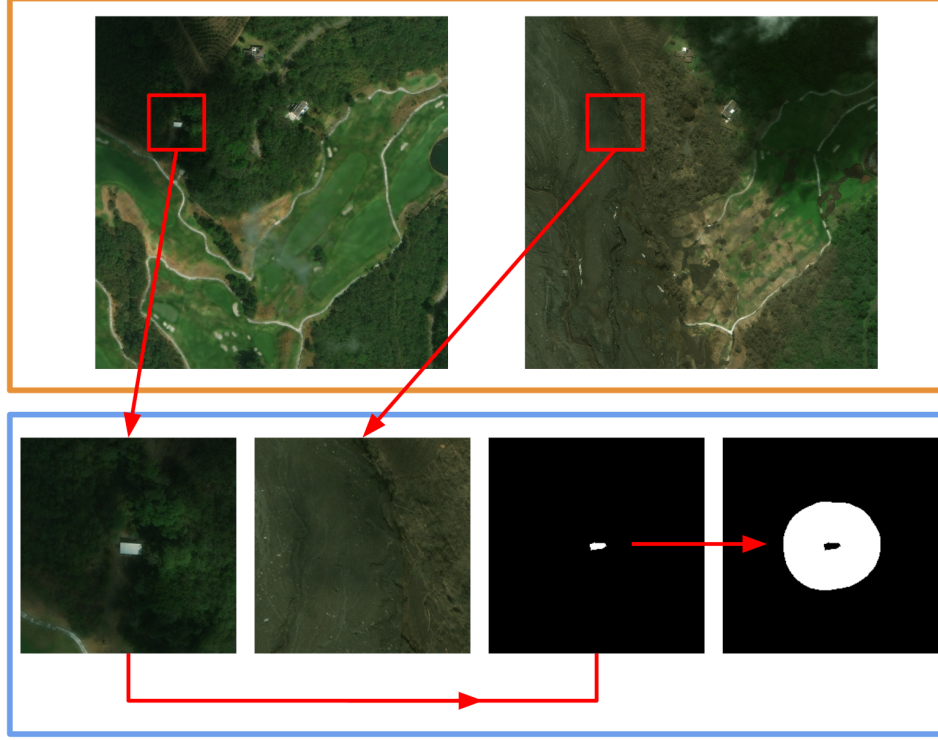


Figure 5: An example profile set for a building in a given remote sensing cell during Guatemala Volcano hazard. From the pre-disaster image (top-left) and the post-disaster image (top-right), a building-centered crop is performed to get pre- and post-crops (bottom-left, first and second) for localized processing. Then, the building polygon is extracted and saved to a mask (bottom-right, second), and a buffer mask of the polygon boundary is also created for spatial context.

The classification head consumes the concatenation $[v^{\text{pre}} \parallel v^{\text{post}} \parallel \Delta v \parallel v^{R,\text{pre}} \parallel v^{R,\text{post}} \parallel \Delta v^R]$ and produces $K-1$ cumulative logits $g_1, \dots, g_{K-1}$ under a CORAL formulation (Cao et al., 2020). With ordinal label $y \in \{0, \dots, K-1\}$, the target vector is $t_k = \mathbb{I}[k \leq y]$. The training loss is the sum of binary cross-entropies over the cumulative tasks,

$$\mathcal{L}_{\text{CORAL}} = \sum_{k=1}^{K-1} \text{BCE}(\sigma(g_k), t_k),$$

and inference converts cumulative probabilities $c_k = \sigma(g_k)$ to class probabilities via

$$p(0) = 1 - c_1, \quad p(k) = c_k - c_{k+1} \quad (1 \leq k \leq K-2), \quad p(K-1) = c_{K-1}.$$

Reported metrics include macro-F1 and per-class F1, Quadratic Weighted Kappa for ordinal agreement, Negative Log-Likelihood, and a tolerant accuracy $\mathbb{I}(|\hat{y} - y| \leq 1)$. Splits are tile-blocked to limit spatial/event leakage.

Uncertainty and calibration. The model outputs the full per-building class-probability vector $p \in [0, 1]^K$. From p we compute confidence and uncertainty summaries, including the maximum class probability $p_{\max} = \max_k p(k)$, the margin $m = p_{(1)} - p_{(2)}$ between the two largest probabilities, Shannon entropy $H(p) = -\sum_k p(k) \log p(k)$, and the expected severity $s = \sum_k k p(k)$. The ordinal variance

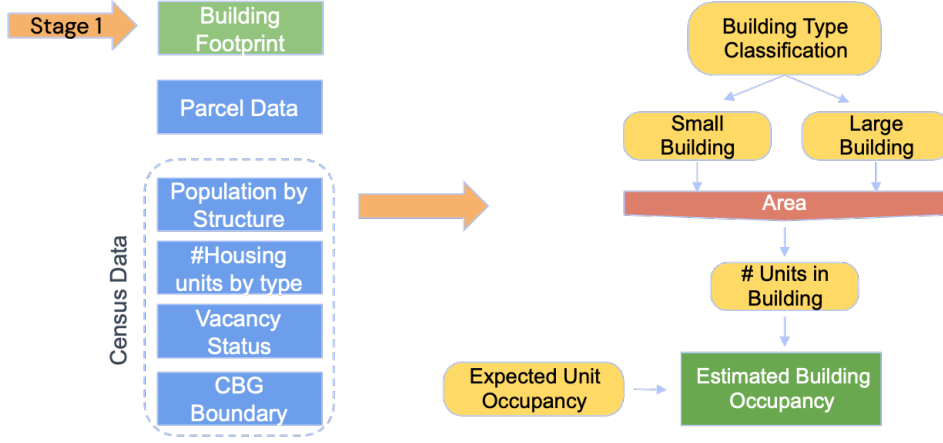


Figure 6: Building footprint estimate from Stage 1 model is taken, and combined with parcel and CBG level attributes to estimate occupancy per structure

$\text{Var}[K] = \sum_k p(k) (k - s)^2$ will be added in upcoming iterations. Reliability is quantified using Expected Calibration Error (ECE) (Guo et al., 2017),

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} |\text{acc}(S_b) - \text{conf}(S_b)|,$$

where S_b indexes probability bins, $\text{acc}(\cdot)$ is empirical accuracy, and $\text{conf}(\cdot)$ is mean predicted confidence in each bin. Two post-hoc calibration options are supported: (i) scalar temperature scaling that divides logits by $T > 0$ chosen on validation to minimize NLL; and (ii) class-wise vector temperature in probability space, $p'_c \propto p_c^{\alpha_c}$ with normalization, where exponents α_c are selected on validation to improve NLL/ECE. At inference time, p , m , H , and s are persisted per building for quality assurance and abstention rules, with optional regional Monte Carlo aggregation for higher-level summaries planned.

2.4 Proxy Rule & Demographic Dataset Coupling

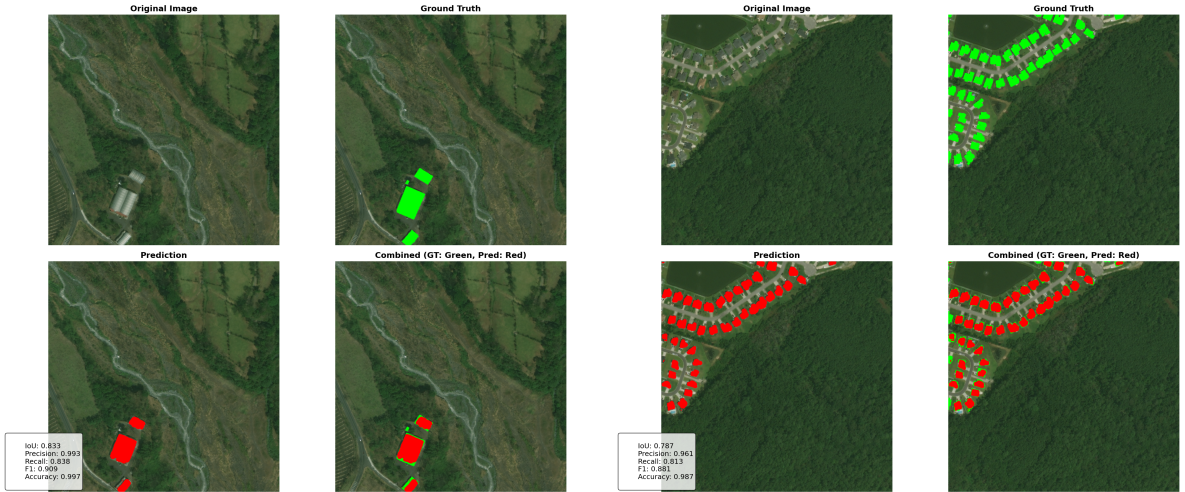
To assign demographic attributes to buildings, we implement a spatial disaggregation framework that assigns population estimates to individual buildings by integrating parcel land-use classifications, building footprints, and Census Block Group (CBG) demographics for Harris County, Texas. Buildings are associated to parcels and CBGs via a two-stage spatial filter: first, a bounding-box elimination, then a centroid-based point-in-polygon test—followed by within-type spatial joins that ensure polygons are correctly nested and located for subsequent analysis. Each building is classified into residential types using parcel land-use codes, with fixed unit counts for single-family, duplex, and triplex structures, and area-based unit estimation for multi-family buildings to reflect varying densities across the housing stock. Area-based thresholds distinguish small and large multi-family structures for unit estimation fidelity, and all geometry processing is conducted in the Texas State Plane Central projection to ensure accurate area calculations prior to re-projecting for web mapping. Population is assigned per building using the equation $\text{Estimated Population} = \text{Estimated Units} \times \text{People Per Unit Ratio} \times \text{Occupancy Rate}$, where ratios are derived at the CBG level from American Community Survey (ACS) cross-tabs by structure type to preserve local context. Data quality safeguards include bounded ratio checks, fallback defaults when ACS inputs are missing or unstable, and population caps by building type to prevent implausible outcomes, all tracked through a tiered quality-flag system that records defaults, caps, unusually large buildings, and other estimation issues. Figure 6 shows the steps followed for this aggregation. Validation aggregates building estimates to CBG totals and evaluates accuracy

against ACS population using mean absolute error, mean absolute percentage error, and correlation metrics, providing transparency on coverage, reliance on defaults, and error distributions.

3 Results

3.1 Stage 1 Model

Qualitative results demonstrate that the model successfully delineates building footprints in diverse environments. Figure 7 shows two representative test samples, including the original pre-disaster image, the ground-truth mask, the predicted mask, and an overlay visualization combining ground-truth (green) and prediction (red). The first example illustrates accurate footprint delineation of isolated buildings, while the second highlights the model’s ability to detect dense housing layouts with some discrepancies at boundaries.



(a) Example 1: Original, Ground Truth, Prediction, and Overlay.

(b) Example 2: Original, Ground Truth, Prediction, and Overlay.

Figure 7: Qualitative segmentation results. Ground-truth is shown in green, predictions in red.

Training dynamics are shown in Figure 8, where both training and validation loss steadily decrease, indicating stable convergence without major overfitting.

Quantitative evaluation is reported in Table 1. The model achieves a mean IoU of 0.521, and mean recall of 0.585, reflecting reasonable balance between false positives and false negatives. The mean F1 score is 0.620, and the mean accuracy is consistently high at 0.977, showing strong performance across test samples.

Metric	Mean	Std. Dev.
IoU	0.521	0.301
Recall	0.585	0.319
F1 Score	0.620	0.332
Accuracy	0.977	0.019

Table 1: Segmentation metrics (threshold = 0.5).

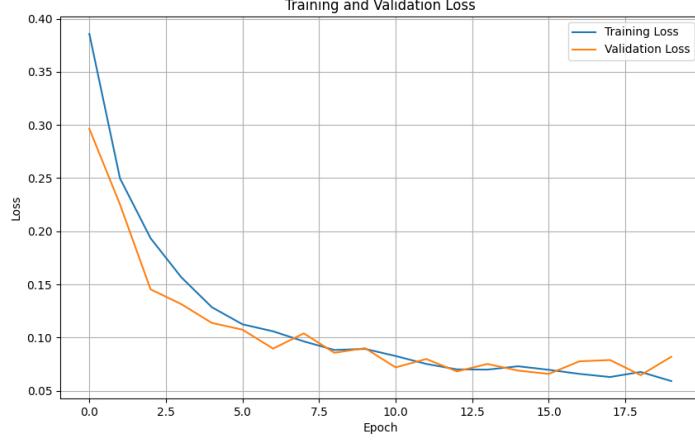


Figure 8: Training and validation loss across epochs.



Figure 9: An example inference provided by the stage 2 model that provides metrics in the upper left corner. The model predicts major disaster damage, has confidence of 0.96, a ground truth of major, a margin of 0.94, and entropy of 0.19. An overall highly confident and correct prediction.

3.2 Stage 2 Model

On a held-out tile-blocked test split, the Siamese CORAL classifier achieves macro-F1 = 0.6029 and QWK = 0.8628 before calibration; after vector temperature calibration, accuracy nudges up to macro-F1 = 0.6081 and QWK = 0.8678, while reliability improves markedly: NLL drops from 0.4631 $\rightarrow$ 0.4106 and ECE falls from 0.4383 to 0.0253/0.0188/0.0290 (10/15/20 bins). Class-wise tolerant accuracy ($|\hat{y} - y| \leq 1$) remains high (e.g., class 0: 0.9936 $\rightarrow$ 0.9930; class 2: 0.9284 $\rightarrow$ 0.9117; class 3: 0.9107 $\rightarrow$ 0.9107), consistent with most confusions occurring between adjacent severities. Importantly, calibration does not change class assignments but does correct confidence, which is why F1/QWK are essentially stable while NLL/ECE improve; this makes per-building probability vectors suitable for downstream expectation-based aggregation and uncertainty reporting. The learned class-wise exponents for vector calibration on validation were [0.6, 1.5, 0.6, 1.2], and we also log per-building confidence diagnostics (top-p, margin, entropy) to support triage and abstention in ambiguous cases.

Inference is provided on the building level, and a visual example is given by Figure 9. The inference provides the prediction, confidence, optional ground truth in training mode, margin, and entropy with a colored overlay on the post-disaster image crop, specified by section 2.3.3 and 2.3.4. The model is still vulnerable to incorrect predictions with both high and low confidence across all damage subtypes, but the high confidence mistakes mostly concentrate on predicting no damage over minor ground truth.

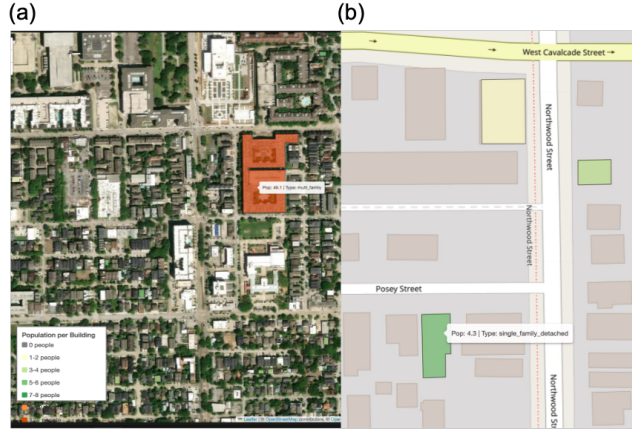


Figure 10: Building occupancy assignment based on proxy rules from coarser demographic data. Panel (a) shows the implementation of model for large residential complex and (b) for single family dwellings

3.3 Proxy Rules

The proxy rule model was implemented for Harris County (Houston metropolitan area), Texas, as a test case. 100,000 building boundaries were randomly sampled from the region to implement the model due to computational constraints. Figure 10 shows the occupancy estimates derived from our model. Based on visual verification, the model accurately represents large and small houses and appropriately assigns building occupancy based on the observed population density observed in the CBG and the building category from the parcel data.

The findings show that the model can perform well, given the absence of availability of such high-resolution data, and with minor tweaks in the thresholds, can be implemented for any region in the US.

4 External Case Study: LA Fire 2025

4.1 Background and Dataset

In January 2025, a series of devastating wildfires struck across Los Angeles, spreading through multiple locations in both dense urban areas and the wildland–urban interface (WUI). The fires caused catastrophic destruction, resulting in an estimated 250 billion USD in damages and heavily impacting communities in Palisades, Pasadena/Eaton, and Malibu. Because the disaster unfolded in multiple zones simultaneously, emergency response efforts faced significant challenges in allocating resources equitably across different affected communities. This competition for firefighting resources, evacuation support, and post-disaster recovery highlighted the difficulties of managing large-scale disasters that span several urbanized areas at once.

For research and validation purposes, we leverage commercial satellite imagery provided by Maxar Technologies, which made its pre- and post-disaster data openly available through the Maxar Open Data Program. These datasets provide critical insights into the spatial extent of destruction, enabling systematic comparisons before and after the fire. Unfortunately, however, the pre-disaster coverage for Palisades and Malibu is missing, and despite more than half a year having passed since the disaster, Maxar has not released the missing archives. This gap poses a major obstacle to comprehensive pre–post change analysis in these zones, leaving large portions of urban area undocumented in the baseline imagery. Figure 2

It is important to emphasize that this dataset is considered external data in our study. The Maxar

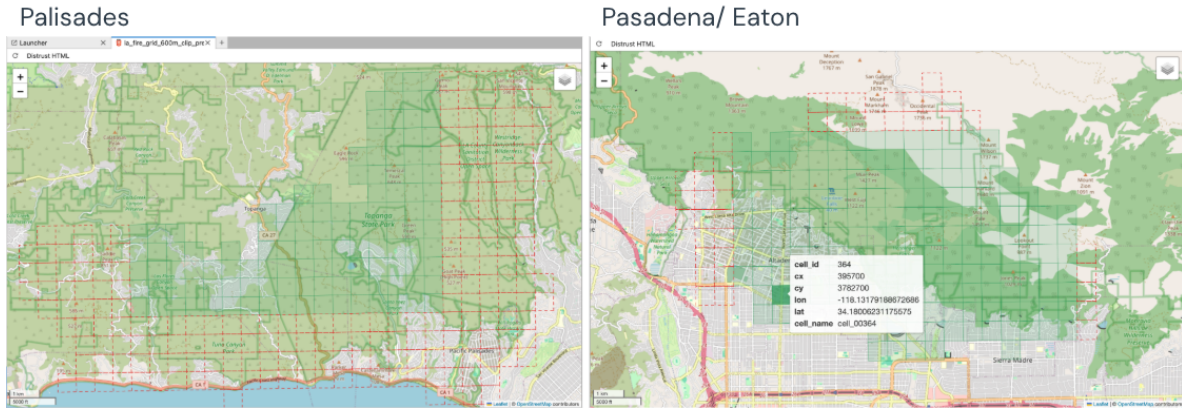


Figure 11: Extent of LA fire in Palisades and Altadena, with MAXAR image availability, Green: Available, Red: Unavailable

imagery was not used for training our model; rather, it is incorporated as an independent case study to test the transferability and robustness of our methods in a real-world disaster response context.

4.2 Methods and Preliminary Results

In this case study, we duplicated our established framework to evaluate the impacts of the 2025 Los Angeles wildfire using Maxar’s open access imagery. The methodology follows two key stages:

Stage 1: Building Footprint Extraction Pre-disaster images were used to generate building footprints across the study area. These serve as the baseline layer to understand the pre-fire urban fabric.

Stage 2: Damage Classification Multiple post-disaster images, collected on different dates, were compared against the pre-fire baseline to classify destruction levels for individual buildings. To account for variability in image quality—such as the presence of smoke, cloud cover, or optical distortions—we aggregate across available dates. The final damage level assigned to each structure corresponds to the mode of predictions across the valid post-event images, reducing bias from single-image anomalies.

Preliminary Results The initial results demonstrate both the promise and current limitations of the approach:

Stage 1: The framework successfully captured the overall patterns and community-level structures, reflecting the urban footprint of affected neighborhoods. However, at the individual housing level, the accuracy of footprint delineation remains limited, with room for refinement to detect smaller or irregular structures more precisely.

Stage 2: At this stage, we are not yet able to produce a city-wide destruction map. Instead, we conducted targeted assessments on selected neighborhoods. In these areas, several buildings were successfully recognized, and their damage levels (no damage, minor, major, or destroyed) were classified with encouraging accuracy. Figure 12 shows that the workflow can identify localized destruction, though further scaling is required.

Future Extensions To build on these results, a third stage will integrate downscaled demographic and community-level information. By linking structural damage with socioeconomic indicators, the framework aims to generate fine-scale destruction and vulnerability maps. These outputs will be

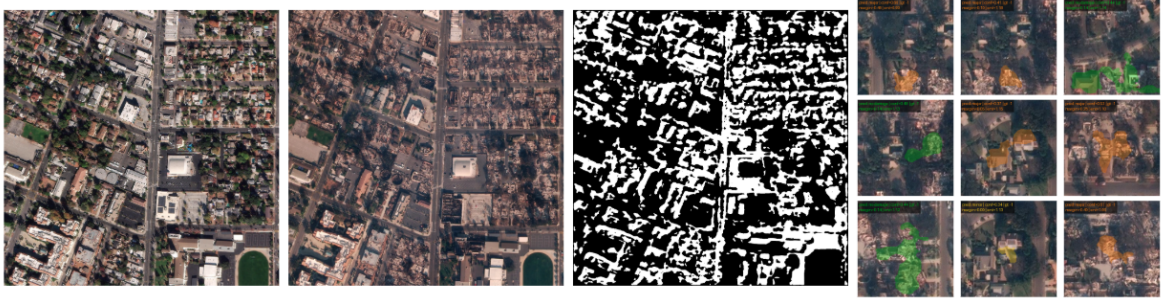


Figure 12: Sample Cell 364, from left to right: Pre, Post, Building Footprint, Damage Level

valuable for first responders and disaster management agencies, enabling rapid assessments of impacts across different communities and supporting equitable resource allocation.

5 Discussion

The two-stage pipeline produces building-level damage estimates with calibrated probabilities suitable for downstream use. On a held-out, tile-blocked test split, Stage 2 attains macro-F1 = 0.6029 and QWK = 0.8628 before calibration; vector temperature calibration leaves class assignments stable while improving reliability. Confusions concentrate between adjacent severities, which aligns with an ordinal formulation and supports reporting tolerant accuracy alongside per-class F1.

Uncertainty tracking adds practical value beyond point predictions. Because the model retains per-building probability vectors, we can expose confidence and ambiguity signals such as top probability, margin between the two most likely classes, entropy, and expected severity, and we can, when desired, summarize uncertainty at neighborhood scales by Monte Carlo draws over the categorical distributions. Early qualitative checks indicate that high-entropy buildings cluster where polygons are ambiguous or post-event imagery degrades, which is an appropriate target for abstention or human review and a natural basis for sensitivity analyses that vary Stage 1 quality and read out Stage 2 probability stability. In parallel, the proxy rule that combines residential filtering with geometry-based unit assignment yields tract-consistent, spatially plausible unit distributions when aggregated, providing a working path from predicted damage to interpretable building-level impact summaries.

Limitations include persistent class imbalance, potential error propagation from Stage 1 to Stage 2, the need for a structured sensitivity study that perturbs Stage 1 quality while tracking Stage 2 uncertainty, and the lack of easily shareable building-level ground truth for validating the proxy rule on public datasets. Future work will explore ways to address these limitations, add street-view coupling to enrich building attributes where available, and run comparative studies against strong one-stage semantic segmentation baselines to quantify the benefits of the instance-first design.

6 Conclusion

This progress report demonstrates a feasible end-to-end framework for building-level disaster impact analysis grounded in an instance-first, two-stage design. Stage 1 generates polygons from pre-disaster imagery that are sufficient to drive geometry-centered profiling at scale. Stage 2 uses a Siamese encoder with an ordinal head to produce calibrated per-building probabilities that support both accuracy and interpretability. The system exports full probability vectors per building together with confidence and ambiguity summaries, enabling threshold-free aggregation and targeted human review. A multi-modal proxy rule couples predicted damage to building-level occupancy estimates and yields tract-consistent, spatially plausible allocations when aggregated.

Current limitations include persistent class imbalance, potential error propagation from Stage 1 to Stage 2, the need for systematic sensitivity analysis that perturbs Stage 1 quality while reading out Stage 2 uncertainty, and restricted access to building-level ground truth for validating the proxy rule. Future work will address these limitations, integrate street-view information to enrich building attributes where available, and run comparative studies against strong one-stage semantic segmentation baselines to quantify the benefits of the instance-first approach. The results to date establish a preliminary foundation for an operational pipeline that delivers building-centric, calibrated, and equity-aware disaster assessments from user-specified areas and time windows.

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References

- [1] Sultan Al Shafian, Chao He, and Da Hu. Damagescope: An integrated pipeline for building damage segmentation, geospatial mapping, and interactive web-based visualization. *Remote Sensing*, 17(13), 2025. ISSN 2072-4292. doi: 10.3390/rs17132267. URL <https://www.mdpi.com/2072-4292/17/13/2267>.
- [2] Irene Alisjahbana, Jiawei Li, Yue Zhang, et al. Deepdamagenet: A two-step deep-learning model for multi-disaster building damage segmentation and classification using satellite imagery. *arXiv preprint arXiv:2405.04800*, 2024.
- [3] Colin Alstad. The xview2 ai challenge, 2020. URL <https://www.ibm.com/think/insights/the-xview2-ai-challenge>. IBM Think Insights.
- [4] Yanbing Bai, Junjie Hu, Jinhua Su, Xing Liu, Haoyu Liu, Xianwen He, Shengwang Meng, Erick Mas, and Shunichi Koshimura. Pyramid pooling module-based semi-siamese network: A benchmark model for assessing building damage from xbd satellite imagery datasets. *Remote Sensing*, 12(24), 2020. ISSN 2072-4292. doi: 10.3390/rs12244055. URL <https://www.mdpi.com/2072-4292/12/24/4055>.
- [5] Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331, 2020. ISSN 0167-8655. doi: <https://doi.org/10.1016/j.patrec.2020.11.008>. URL <https://www.sciencedirect.com/science/article/pii/S016786552030413X>.
- [6] Liwei Deng and Yue Wang. Post-disaster building damage assessment based on improved u-net. *Scientific reports*, 12(1):15862, 2022.
- [7] Zhipeng Ding, Xu Han, Peirong Liu, and Marc Niethammer. Local temperature scaling for probability calibration. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6889–6899, 2021.
- [8] Marios Galanis, Krishna Rao, Xinle Yao, Yi-Lin Tsai, Jonathan Ventura, and G. Andrew Fricker. Damagemap: A post-wildfire damaged buildings classifier. *International Journal of Disaster Risk Reduction*, 65:102540, 2021. ISSN 2212-4209. doi: <https://doi.org/>

- 10.1016/j.ijdr.2021.102540. URL <https://www.sciencedirect.com/science/article/pii/S221242092100501X>.
- [9] Tia Sherée Gaynor and Meghan E. Wilson. Social vulnerability and equity: The disproportionate impact of covid-19, Aug 2020. URL https://www.researchgate.net/publication/342372036_Social_Vulnerability_and_Equity_The_Disproportionate_Impact_of_COVID-19.
- [10] Shahrzad Gholami, Caleb Robinson, Anthony Ortiz, Siyu Yang, Jacopo Margutti, Cameron Birge, Rahul Dodhia, and Juan Lavista Ferres. On the deployment of post-disaster building damage assessment tools using satellite imagery: A deep learning approach. In *2022 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 1029–1036. IEEE, 2022.
- [11] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.
- [12] Ritwik Gupta, Richard Hosfelt, Sandra Sajeev, Nirav Patel, Bryce Goodman, Jigar Doshi, Eric Heim, Howie Choset, and Matthew Gaston. xbd: A dataset for assessing building damage from satellite imagery, 2019. URL <https://arxiv.org/abs/1911.09296>.
- [13] Victor Hertel, Omar Wani, Christian Geiß, Marc Wieland, and Hannes Taubenböck. Bayesiammtl: Uncertainty-aware multitask learning for post-disaster building damage assessment. *International Journal of Applied Earth Observation and Geoinformation*, 143:104759, 2025.
- [14] Hoesung Lee, Katherine Calvin, Dipak Dasgupta, Gerhard Krinner, Aditi Mukherji, Peter Thorne, Christopher Trisos, José Romero, Paulina Aldunce, Ko Barret, et al. *ipcc, 2023: Climate change 2023: Synthesis report, summary for policymakers. contribution of working groups i, ii and iii to the sixth assessment report of the intergovernmental panel on climate change [core writing team, h. lee and j. romero (eds.)]. ipcc, geneva, switzerland. 2023.*
- [15] Cheng Liao, Han Hu, Haifeng Li, Xuming Ge, Min Chen, Chuangnong Li, and Qing Zhu. Joint learning of contour and structure for boundary-preserved building extraction. *Remote Sensing*, 13(6):1049, 2021.
- [16] Jinhua Su, Yanbing Bai, Xingrui Wang, Dong Lu, Bo Zhao, Hanfang Yang, Erick Mas, and Shunichi Koshimura. Technical solution discussion for key challenges of operational convolutional neural network-based building-damage assessment from satellite imagery: Perspective from benchmark xbd dataset. *Remote Sensing*, 12(22), 2020. ISSN 2072-4292. doi: 10.3390/rs12223808. URL <https://www.mdpi.com/2072-4292/12/22/3808>.
- [17] UNITAR. Unosat introduces ai in its flood rapid mapping operations for the benefit of national disaster management authorities., Aug 2020. URL <https://unitar.org/about/news-stories/news/unosat-introduces-ai-its-flood-rapid-mapping-operations-benefit-national-disaster-manag>
- [18] Fabien H. Wagner, Ricardo Dalagnol, Yuliya Tarabalka, Tassiana Y. F. Segantine, Rogério Thomé, and Mayumi C. M. Hirye. U-net-id, an instance segmentation model for building extraction from satellite images—case study in the joanópolis city, brazil. *Remote Sensing*, 12(10), 2020. ISSN 2072-4292. doi: 10.3390/rs12101544. URL <https://www.mdpi.com/2072-4292/12/10/1544>.